

141 functions #2

clicker questions

Suppose you are writing test inputs for Codingbat

For the next question, pretend that you are writing the test inputs for Codingbat

Students are writing a function that has three parameters (a, b, c) and returns true if *at least* one parameter is positive, and false otherwise.

```
public static boolean onePos(int a, int b, int c)
```

You are testing 3, 4, 5 as an input (it returns true).

Which additional input is *least* helpful?

A	0, 0, 0
B	-10, -10, -10
C	1, 100, 100000
D	1, 0, 1
E	-1, 0, 1

Students are writing a function that has three parameters (a, b, c) and returns true if *at least* one parameter is positive, and false otherwise.

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public static boolean onePos(int a, int b, int c)
```

You are testing 3, 4, 5 as an input (it returns true).

Which additional input is *least* helpful?

A	0, 0, 0
B	-10, -10, -10
C	1, 100, 100000
D	1, 0, 1
E	-1, 0, 1

This doesn't tell you anything new that 3,4,5 would not also determine. All of the other inputs test conditions that are more likely to catch errors!

Which is a more useful and flexible function?

A

```
public static boolean isEven(int num) {  
    return num % 2 == 0;  
}
```

not A

```
public static void isEven() {  
    Scanner sc=new Scanner(System.in);  
    int num=sc.nextInt();  
    System.out.println(num % 2 == 0);  
}
```

Which is a more useful and flexible function?

A

```
public static boolean isEven(int num) {  
    return num % 2 == 0;  
}
```

not

A

```
public static void isEven() {  
    Scanner sc=new Scanner(System.in);  
    int num=sc.nextInt();  
    System.out.println(num % 2 == 0);  
}
```

This function is flexible and more useful!

```
public static boolean isEven(int num) {  
    return num % 2 == 0;  
}
```

The parameter is an int and returns a boolean. You can plug this function in wherever you need it!

This function is not very flexible

```
public static void isEven() {  
    Scanner sc=new Scanner(System.in);  
    int num=sc.nextInt();  
    System.out.println(num % 2 == 0);  
}
```

You need someone at the keyboard to make this work, and someone needs to read the printing. It would be hard to combine this function with other functions.

```
public static String mystery(String s) {  
    // put empty string in x  
    String x = "";  
    // write a for loop with:  
    // initializer i=0, update i++  
    // condition i<s.length()  
    for (int i = 0; i < s.length(); i++) {  
        // assign s.charAt(i)+x into x  
        x = s.charAt(i) + x;  
    }  
    // return the value x from the function  
    return x;  
}
```

How good are these comments, from A (great) to E (terrible).

```
public static String mystery(String s) {  
    // put empty string in x  
    String x = "";  
    // write a for loop with:  
    // initializer i=0, update i++  
    // condition i<s.length()  
    for (int i = 0; i < s.length(); i++) {  
        // assign s.charAt(i)+x into x  
        x = s.charAt(i) + x;  
    }  
    // return the value x from the function  
    return x;  
}
```

These comments make no effort to describe the overall algorithm being used.

Anyone who knows Java knows what x="" does!

“what this code does” is CQ on next slide!

D or E: These comments are pretty useless. They just re-state what each line of code does.

What does this return?

```
public static String mystery(String s) {  
    String x = "";  
    for (int i = 0; i < s.length(); i++) {  
        x = s.charAt(i) + x;  
    }  
    return x;  
}
```

- A. the empty string
- B. the number of occurrences of 'x'
- C. s (i.e. s remains unchanged)
- D. s with the characters reversed
- E. true if s has an even number of characters; false otherwise

What does this return?

```
public static String mystery(String s) {  
    String x = "";  
    for (int i = 0; i < s.length(); i++) {  
        x = s.charAt(i) + x;  
    }  
    return x;  
}
```

- A. the empty string
- B. the number of occurrences of 'x'
- C. s (i.e. s remains unchanged)
- D. s with the characters reversed**
- E. true if s has an even number of characters; false otherwise

```
public static String mystery(String s) {  
    String x = "";  
    // Reverse the string by going through  
    // each character, but then  
    // concatenate each character  
    // before the chars we've already seen  
    for (int i = 0; i < s.length(); i++) {  
        x = s.charAt(i) + x;  
    }  
    return x;  
}
```

These comments are more useful! We are explaining the difficult parts of the code.

What does this print?

```
public static void go(int x) {  
    x+=5;  
}  
  
public static void main(String[] args) {  
    int z=10;  
    go(z);  
    System.out.println(z);  
}
```

A: 5

B: 10

C: 15

D: 0

E: none of the above

What does this print?

```
public static void go(int x) {  
    x+=5;  
}  
  
public static void main(String[] args) {  
    int z=10;  
    go(z);  
    System.out.println(z);  
}
```

A: 5

B: 10

C: 15

D: 0

E: none of the above

Draw primitive/reference diagram
for pass by value/ pass by reference

Convert our F2C program to a function

What are the inputs? What are their types?

What is the output? What is the return type?

What should we name the function?

```
public static double f2c(double f)
{
    double ctemp = (f-32) * 5.0/9;
    return ctemp;
}
```

```
public static double f2c(double f)
{
    double ctemp = (f-32) * 5.0/9;
    return ctemp;
}
```

Can you make the body have only one line?

A: yes

not A: no

```
public static double f2c(double f)
{
    double ctemp=(f-32) * 5.0/9;
    return ctemp;
    return (f-32) * 5.0/9;
}
```

Can you make the body have only one line?

A: yes

not A: no

Which correctly invokes this function?

```
public static int sumN(int n) {  
    int sum = 0;  
    for (int i = 1; i <= n; i++) {  
        sum += i;  
    }  
    return sum;  
}
```

sumN(5)	A
sumN(int 5)	B
sumN(sumN(5))	C
sumN()	D
not exactly one of the above	E

Which correctly invokes this function?

```
public static int sumN(int n) {  
    int sum = 0;  
    for (int i = 1; i <= n; i++) {  
        sum += i;  
    }  
    return sum;  
}
```

A invokes the function with a parameter of 5.

C invokes the function twice! Once with 5, and once with the result of invoking it with 5.

sumN(5)	A
sumN(int 5)	B
sumN(sumN(5))	C
sumN()	D
not exactly one of the above	E

What string would make this return false?

```
public static boolean mystery(String str) {  
    for (int i=0; i<str.length()/2; i++) {  
        if(str.charAt(i) != str.charAt(str.length()-1-i))  
        {  
            return false;  
        }  
    }  
    return true;  
}
```

- A. "a"
- B. "abcba"
- C. "abcabc"
- D. "aaaaaa"
- E. none of the above

What string would make this return false?

```
public static boolean mystery(String str) {  
    for (int i = 0; i < str.length()/2; i++) {  
        if (str.charAt(i) != str.charAt(str.length() - i - 1))  
        {  
            return false;  
        }  
    }  
    return true;  
}
```

- A. "a"
- B. "abcba"
- C. "abcabc"**
- D. "aaaaaa"
- E. none of the above

This checks if the given String is a Palindrome (i.e. the same forward as backward)

Done!

extra slides follow here